



Computer Science & IT



Data Structure & Programming

Graph & Hashing

Lecture No. 02



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Recap of Previous Lecture



Topic

Hashing

Topic

Collision

Topic

Seperate chaining

Topic

Linear probing

Topic

Topics to be Covered



Topic

Quadratic probing

Topic

Double Hashing

Topic

Topic

Topic

Graph Representation

BFS Algorithm



Topic : Hashing



till 2025 All
problem Related

Decidability — PDF

Table

Decidability undecidability — proof oriented



Halting problem

Reducibility
2 Theorem

Undecidable $\leq$ Unknown

Unknown problem has to
be undecidable

Unknown $\leq$ D

Unknown is Decidable

L_1 is CFG L_2 CFG $\rightarrow$ closure property
is $L_1 \cup L_2$ is Ambiguous?

Decidable / undecidable



Topic : Hashing



Linear probing suffers from primary cluster problem

primary cluster: Long run of filled slot

in case of collision $x, x+1, x+2, x+3$



Topic : *Primary Clustering*

Primary clustering is the tendency for a collision resolution scheme such as linear probing to create *long runs of filled slots* near the hash position of keys.

If the primary hash index is x , subsequent probes go to $x + 1$, $x + 2$, $x + 3$ and so on, this results in Primary Clustering.

Once the *primary cluster forms*, the bigger the cluster gets, the *faster it grows*. And it *reduces the performance*.



Topic : Hashing



Quadratic probing

$$\text{probe}_0 = h(k)$$

$$\text{probe}_1 = (h(k) + 1^2) \% m$$

$$\text{probe}_2 = (h(k) + 2^2) \% m$$

$$\text{probe}_3 = (h(k) + 3^2) \% m$$

$$\text{probe}_i = (h(k) + i^2) \% m$$

Quadratic Probing

Suppose that an element should appear in bucket h :

if bucket h is occupied, then check the following sequence of bucket:

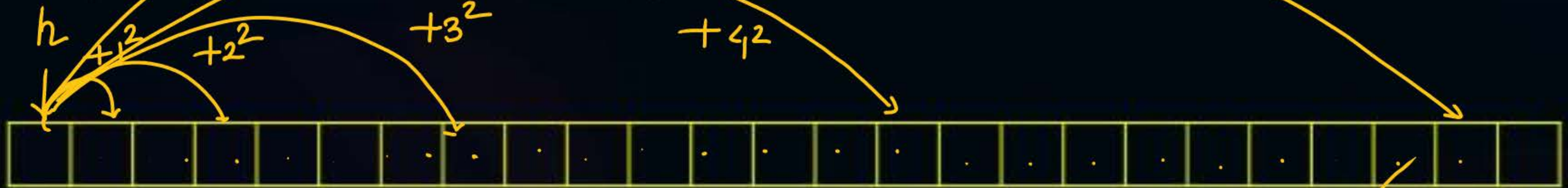
$$h + 1^2, h + 2^2, h + 3^2, h + 4^2, h + 5^2,$$

...

$$h + 1, h + 4, h + 9, h + 16, h + 25, \dots$$

Quadratic Probing

If one of $h + i^2$ falls into a cluster, this does not imply the next one to map into position i will also belong to the cluster.



Example Quadratic Probing

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

Key - x	$h(x, i) = (h(x), i^2) \bmod 20$
96	$96 \% 20 = 16$
48	$48 \% 20 = 8$
63	$63 \% 20 = 3$
29	$29 \% 20 = 9$
87	$87 \% 20 = 7$
77	$77 \% 20 = 17$
47	$47 \% 20 = 7$
73	
70	
86	
166	

$$\begin{aligned} \text{probe}_0 &= 47 \% 20 = 7 \\ \text{probe}_1 &= (7 + 1^2) \% 20 \\ &= 8 \\ \text{probe}_2 &= (7 + 2^2) \% 20 \end{aligned}$$

	0
	1
	2
63	3
	4
	5
	6
87	7
48 ✓	8
29	9
	10
47	11
	12
	13
	14
	15
96	16
77	17
	18
	19

Example Quadratic Probing

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

Key - x	$h(x, i) = (h(x), i^2) \bmod 20$
96	$96 \% 20 = 16$
48	$48 \% 20 = 8$
63	$63 \% 20 = 3$
29	$29 \% 20 = 9$
87	$87 \% 20 = 7$
77	$77 \% 20 = 17$
47	$47 \% 20 = 7$
73	$73 \% 20 = 13$
70	$70 \% 20 = 10$
86	$86 \% 20 = 6$
166	$166 \% 20 = 6$

$$\text{probe}_0 = 166 \% 20 = 6$$

$$\text{probe}_1 = (6 + 1^2) \% 20 = 7$$

$$\text{probe}_2 = (6 + 2^2) \% 20 = 10$$

$$\text{probe}_3 = (6 + 3^2) \% 20 = \underline{15}$$

	0
	1
	2
63	3
	4
	5
86	6
87	7
48 ✓	8
29	9
70	10
47	11
	12
73	13
	14
166	15
96	16
77	17
	18
	19

Example Quadratic Probing

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

Key - x	$h(x, i) = (h(x) + i^2) \bmod 20$
96	$96 \% 20 = 16$
48	$48 \% 20 = 8$
63	$63 \% 20 = 3$
29	$29 \% 20 = 9$
87	$87 \% 20 = 7$
77	$77 \% 20 = 17$
47	$47 \% 20 = 7$
73	$73 \% 20 = 13$
70	$70 \% 20 = 10$
86	$86 \% 20 = 6$
166	$166 \% 20 = 6$

Insert key 206

$$\text{probe}_0 = 206 / 20 = 10$$

$$\text{probe}_1 = (10 + 1^2) \% 20 = 11$$

$$\text{probe}_2 = (10 + 2^2) \% 20 = 14$$

$$\text{probe}_3 = (10 + 3^2) \% 20 = 19$$

$$\text{probe}_4 = (10 + 4^2) \% 20 = 6$$

	0
	1
	2
63	3
	4
	5
86	6
87	7
48 ✓	8
29	9
70	10
47	11
	12
73	13
	14
166	15
96	16
77	17
	18
	19



Topic : Hashing



Quadratic probe

Insertion of
12 cycle
through the
list and
not found
any place

Key	$h(x, i) = (h(x) + i^2) \bmod 10$
2	$2 \% 10 = 2$
3	$3 \% 10 = 3$
6	$6 \% 10 = 6$
8	$8 \% 10 = 8$
7	$7 \% 10 = 7$
11	$11 \% 10 = 1$
12	$12 \% 10 = 2$

$probe_0 = 12 \% 10 = 2$
 $probe_1 = (2 + 1^2) \% 10 = 3$
 $probe_2 = (2 + 2^2) \% 10 = 6$
 $probe_3 = (2 + 3^2) \% 10 = 1$
 $probe_4 = (2 + 4^2) \% 10 = 8$
 $probe_5 = (2 + 5^2) \% 10 = 7$
 $probe_6 = (2 + 6^2) \% 10 = 8$

	0
11	1
2	2
3	3
	4
	5
6	6
7	7
8	8
	9



Topic : Hashing



Quadratic probing forms scattered cluster (Secondary cluster)
and (II) Insertion of some key may cycle through the list



Topic : Hashing



Double Hashing · In Double Hashing two hash functions are present

Linear
probing offset
was + 1

1 primary hash function $h_1(k) = \text{probe}_0$

2 Secondary hash function $h_2(k) = \text{offset}$

$$\text{probe}_0 = h_1(k)$$

$$\text{probe}_1 = (h_1(k) + \text{offset}) \% m$$

$$\text{probe}_2 = (h_1(k) + 2 \times \text{offset}) \% m$$

$$\text{probe}_3 = (h_1(k) + 3 \times \text{offset}) \% m$$

$$\text{probe}_i =$$

$$(h_1(k) + i \cdot h_2(k)) \% m$$



Topic : Hashing



Key - x	$h_1(x) = x \bmod 13$	$h_2(x) = 8 - (x \bmod 8)$
18	$18 \% 13 = 5$	
41	$41 \% 13 = 2$	
22	$22 \% 13 = 9$	
44	$44 \% 13 = 5$	$8 - (44 \% 8) = 4$
59	$59 \% 13 = 7$	
32		
31		
73		

44	0
	1
41	2
	3
	4
18	5
	6
59	7
	8
22	9
	10
	11
	12

$$\text{probe}_0 = 5$$

$$\text{probe}_1 = (5 + 1 \times 4) \% 13 \\ = 9$$

$$\text{probe}_2 = (5 + 2 \times 4) \% 13 \\ = 13 \% 13 = 0$$



Topic : Hashing



Insert 60
and find
Index

Key - x	$h_1(x) = x \bmod 13$	$h_2(x) = 8 - (x \bmod 8)$
18	$18 \% 13 = 5$	
41	$41 \% 13 = 2$	
22	$22 \% 13 = 9$	
44	$44 \% 13 = 5$	$8 - (44 \% 8) = \underline{4}$
59	$59 \% 13 = 7$	
32	$32 \% 13 = 6$	
31	$31 \% 13 = 5$	$8 - (31 \% 8) = 8 - 7 = 1$
73	$73 \% 13 = 8$	$8 - (73 \% 8) = 8 - 1 = 7$

44	0
	1
41	2
73	3
	4
18	5
32	6
59	7
31	8
22	9
	10
	11
	12

for 31 offset is 1

hence Linear probing

$$\text{probe}_1 = (8 + 7) \% 13 = 2$$

$$\text{probe}_2 = (8 + 2 \times 7) \% 13$$
$$= \underline{9}$$

$$\text{probe}_3 = (8 + 3 \times 7) \% 13$$
$$= 29 \% 13$$
$$= 3$$



Topic : Hashing



Insert 60
and find
Index

Key - x	$h_1(x) = x \bmod 13$	$h_2(x) = 8 - (x \bmod 8)$
18	$18 \% 13 = 5$	
41	$41 \% 13 = 2$	
22	$22 \% 13 = 9$	
44	$44 \% 13 = 5$	$8 - (44 \% 8) = 4$
59	$59 \% 13 = 7$	
32	$32 \% 13 = 6$	
31	$31 \% 13 = 5$	$8 - (31 \% 8) = 8 - 7 = 1$
73	$73 \% 13 = 8$	$8 - (73 \% 8) = 8 - 1 = 7$

44	0
	1
41	2
73	3
	4
18	5
32	6
59	7
31	8
22	9
	10
	11
60	12

$$\text{probe}_0 = 60 \% 13 = 8$$

$$\text{offset} = 8 - (60 \% 8) \\ = 8 - 4 = 4$$

$$\text{probe}_1 = (8 + 4) \% 13 = 12$$

= 12 empty



Topic : Hashing



PYQ-2023

Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 90$ is

$$h_1(k) = k \bmod 23 = 90 \% 23 = 21$$

$$h_2(k) = 1 + (k \bmod 19) = 1 + 90 \% 19 = 1 + 14 = 15$$

$$\begin{aligned} \text{probe}_1 &= (21 + 1 \times 15) \% 23 \\ &= (36) \% 23 = 13 \end{aligned}$$



Topic : Hashing



Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 10$, and the secondary hash function is $h_2(k) = 7 - (k \bmod 7)$. Assume that the table size is 10. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 49$ is

$$\text{probe}_0 = h_1(k) = k \bmod 10 = 49 \% 10 = \underline{9}$$

$$h_2(k) = 7 - (49 \% 7) = 7 - 0 = 7$$

$$\begin{aligned} \text{probe}_1 &= (9 + 1 \times 7) \% 10 \\ &= (16 \% 10) = 6 \end{aligned}$$



Topic : Hashing



Last year PYQ

#Q. In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i -th probe in the open address table is $[h_1(k) + i \cdot h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____.
(Answer in integer)

0	1	2	3	4	5	6	7	8	9	10
	67	79	25			44		63		24

$$\text{probe}_0 = 24 \cdot 11 = 2$$

$$\text{offset} = (1 + 24 \cdot 7) = 1 + 3 = 4$$

$$\text{probe}_1 = (2 + 4) \cdot 11 = 6$$

$$\text{probe}_2 = (2 + 2 \times 4) \cdot 11 = 10$$

THANK - YOU

Graph

Representation

BFS

Tomorrow agenda

Mid square, folding
Method Tomorrow